

Date 2010/12/28
Class BSc 2nd Year
Roll No 68
Shift Day

Physics Practical Sheet
Prithvi Narayan Campus

Experiment No 8
Group P₂
Sub Physics
Set 1

Object of the Experiment (Block Letters) : TO DETERMINE THE WAVELENGTH OF A SODIUM LIGHT USING NEWTON'S RING APPARATUS.

APPARATUS REQUIRED:

- ① A travelling microscope
- ② A sodium lamp
- ③ Newton's ring apparatus
- ④ A spherometer
- ⑤ A convex lens of short focal length

THEORY:

Circular interference fringes produced by enclosing thin air film varying thickness between the surface of a convex lens of large radius of curvature and a plane glass plate are known as Newton's ring. The wavelength of monochromatic light which produce these ring is given by;

$$\lambda = \frac{D_m^2 - D_n^2}{4(m-n)R}$$

Here, 'R' is the radius of curvature of the surface of the lens in contact with the glass plate, D_m and D_n are the diameter of the m th and n th dark or bright rings.

OBSERVATION:

Pitch of the spherometer = 1 mm

No. of circular divisions = 100

$$\text{Least count (L.C)} = \frac{\text{Pitch}}{\text{No. of circular divisions}} = \frac{1}{100} \text{ mm}$$

$$\text{L.C} = 0.01 \text{ mm}$$

Radius of curvature of the surface of the lens (R) = 96 cm

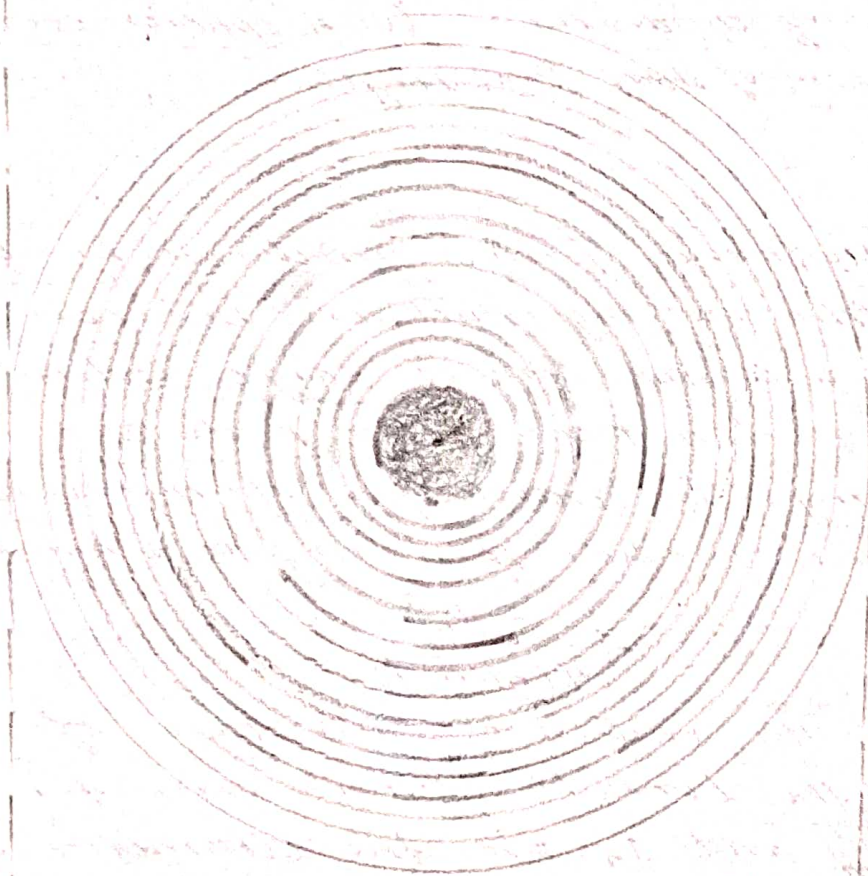


Fig: Newton's ring

OBSERVATION TABLE:

S.N	Ring Number	Micrometer Reading						Diameter	
		Left			Right			$D = a - b$	
		M.S	C.S	Total (a)	M.S	C.S	Total (b)	D in mm	D in cm
1.	4	25	59	25.54	22	37	22.37	2.17	0.317
2.	8	26	24	26.24	22	78	22.78	3.46	0.436
3	12	26	62	26.61	22	33	22.33	5.28	0.528
4	16	26	99	26.99	20	94	20.94	6.05	0.605

CALCULATION:

we have,

$$\lambda = \frac{D_m^2 - D_n^2}{4(m-n)R}$$

For $m=8, n=4$,

$$\begin{aligned} \lambda_1(4,8) &= \frac{D_8^2 - D_4^2}{4(8-4) \times 96} = \frac{(0.436)^2 - (0.317)^2}{16 \times 96} \\ &= 5.8337 \times 10^{-5} \text{ cm} \\ &= 5833 \times 10^{-8} \text{ cm} \\ &= 5833 \times 10^{-10} \text{ m} \\ &= 5833 \text{ \AA} \end{aligned}$$

For $m=12, n=8$,

$$\begin{aligned} \lambda_2(8,12) &= \frac{D_{12}^2 - D_8^2}{4(12-8) \times 96} = \frac{(0.528)^2 - (0.436)^2}{4 \times 4 \times 96} \\ &= 5.7739 \times 10^{-5} \text{ cm} \\ &= 5773 \times 10^{-8} \text{ cm} \\ &= 5773 \times 10^{-10} \text{ m} \\ &= 5773 \text{ \AA} \end{aligned}$$

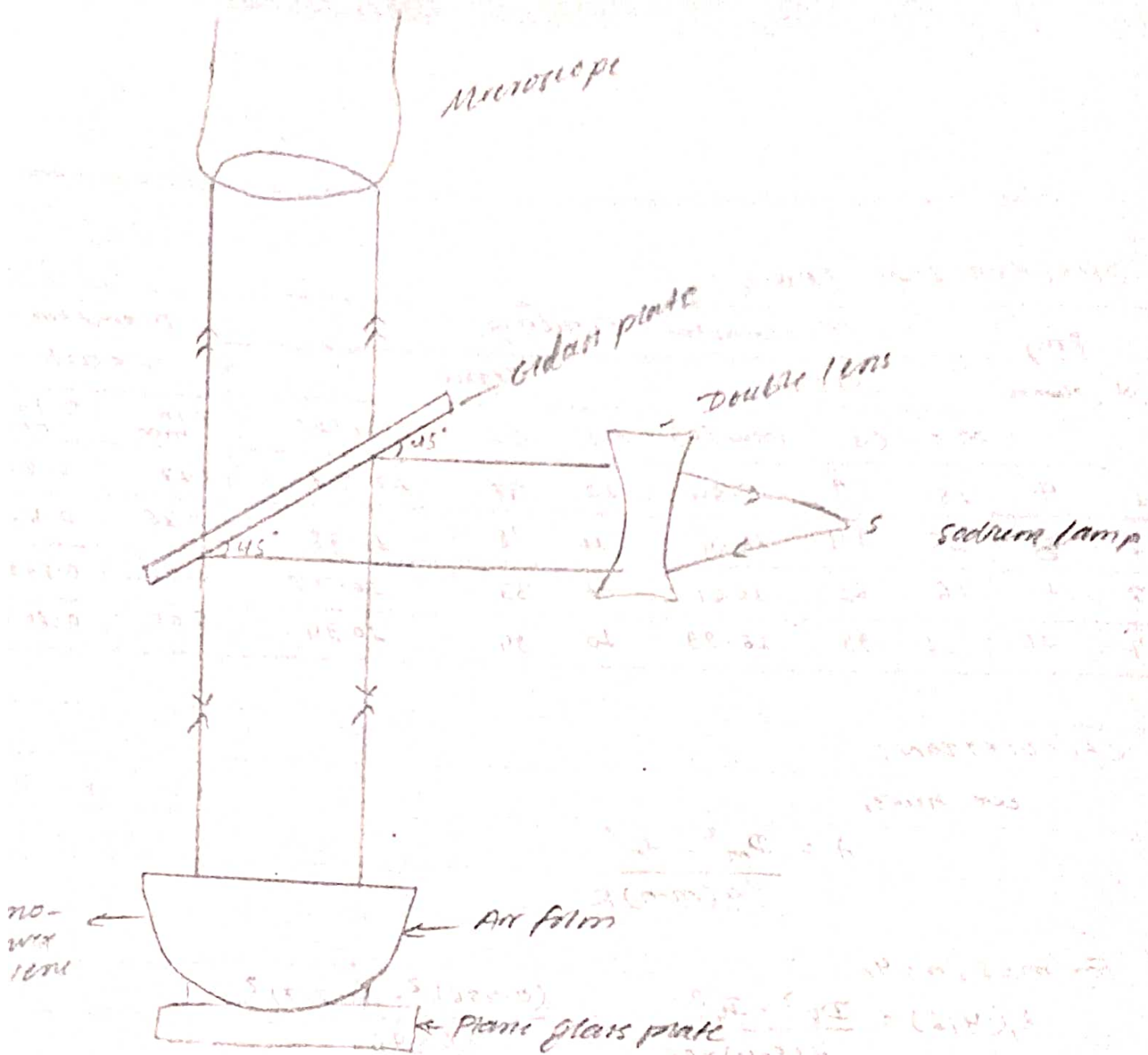


Fig: Interference

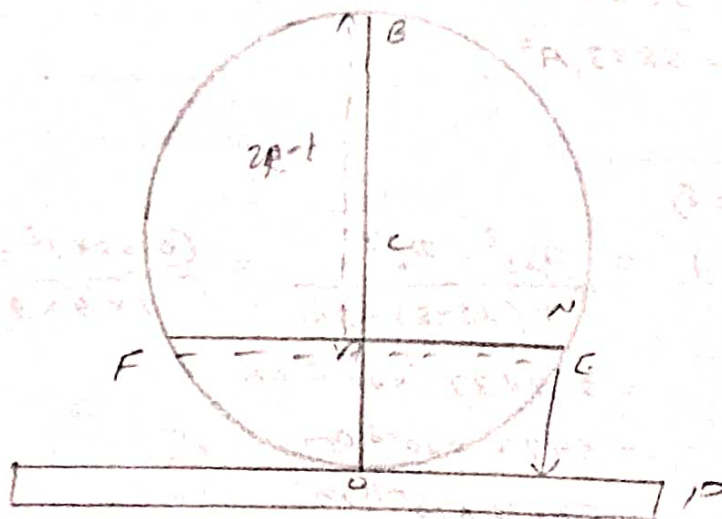


Fig: Phase change at reflection

For $m = 16$, $n = 12$,

$$\begin{aligned}\lambda_3(12, 16) &= \frac{D_{16}^2 - D_{12}^2}{4(16-12) \times 96} = \frac{(0.605)^2 - (0.528)^2}{4 \times 4 \times 96} \\&= 5.6797 \times 10^{-5} \text{ cm} \\&= 5679 \times 10^{-8} \text{ cm} \\&= 5679 \times 10^{-10} \text{ m} \\&= 5679 \text{ \AA}\end{aligned}$$

$$\begin{aligned}\lambda_{\text{com}}(\lambda) &= \frac{\lambda_1 + \lambda_2 + \lambda_3}{3} = \frac{(5833 + 5773 + 5679) \text{ \AA}}{3} \\ \lambda &= 5761.67 \text{ \AA}\end{aligned}$$

ERROR %:

$$\begin{aligned}\text{error \%} &= \left| \frac{\text{S.D value} - \text{calculated value}}{\text{standard value}} \right| \times 100\% \\&= \left| \frac{5893 \text{ \AA} - 5761.67 \text{ \AA}}{5893 \text{ \AA}} \right| \times 100\% \\&= 0.022 \times 100\% \\&= 2.22 \%\end{aligned}$$

RESULT:

Hence, the wavelength of the sodium light was found to be $5761.67 \text{ \AA}$ using Newton's ring apparatus with 2.22% error.

PRECAUTIONS:

- ① The lens and the glass plate should be clean.
- ② A lens of large radius of curvature, perfectly

planconvex should be used.

- (ii) The point of intersection of the cross wires should coincide with the centre of the ring system.
- (iii) The micrometer screw should always be moved in the same direction to avoid error due to back lash.
- (iv) The radius of curvature of the surface of the lens in contact with the glass plate should be measured accurately.